Two Regimes of Metastatic Organotropism: Venous Routing versus Molecular Colonization

Exploratory hypothesis note. The contribution is a conceptual framework and one falsifiable prediction, not a validated predictive model; scope and limitations are stated in full in §9.

1 Introduction and biological context

Metastatic organotropism—the reproducible tendency of each primary cancer to colonize particular distant organs—has been debated since Paget's 1889 "seed and soil" hypothesis, which held that disseminating tumor cells (the "seed") grow only where the microenvironment (the "soil") is compatible. The competing view, associated with Ewing, held that organotropism reflects the mechanical routing of tumor cells by the circulation: cells lodge wherever blood or lymph carries them first. A century of evidence supports *both* mechanisms, and the field treats them as complementary. A relevant third fact, often underweighted, is metastatic inefficiency: the large majority of tumor cells reaching an organ never colonize it [1], so *arrival*, *retention*, and *colonization* are distinct steps that need not share a dominant driver—indeed, the canonical statement of the field is that mechanical factors govern delivery while molecular interactions govern the probability of growth [1].

What has been missing is a principled account of *when* each mechanism dominates. Some tropisms follow vascular anatomy with striking fidelity: prostate carcinoma seeds the axial skeleton through Batson's valveless vertebral venous plexus [2], and colorectal carcinoma seeds the liver through the portal vein. Others track molecular biology instead: within breast cancer, the choice between bone and liver depends on receptor subtype (estrogen-receptor [ER] and human epidermal growth factor receptor 2 [HER2] status), via colonization programs such as ER-regulated SCUBE2 signaling that primes the osteoblast niche [3].

Contribution and its limits. This note makes one conceptual proposal and tests one prediction. The proposal is that these behaviors are two *regimes* of a single process—separable along the arrival/retention/colonization axis above—distinguished by whether the primary's venous drainage is dominated by a single first-pass conduit. We formalize the intuition with a deliberately minimal compartment calculation whose role is to make the regime distinction explicit and to generate predictions; it is not itself a quantitative predictor (see §3, §9). We state this once here and do not re-hedge throughout.

Abbreviations

BBB, blood–brain barrier; CTC, circulating tumor cell; ER, estrogen receptor; HER2, human epidermal growth factor receptor 2; HR, hormone receptor; IVC, inferior vena cava; RCC, renal cell carcinoma; SEER, Surveillance, Epidemiology, and End Results (program); TNBC, triple-negative breast cancer.

2 Core claim and the dominance criterion

Metastatic organotropism is proposed to operate in two regimes:

- **Routing-dominated regime.** A single venous first-pass conduit carries most disseminating cells to one bed; destination is set by venous topology, and molecular seed-and-soil effects are secondary. Examples: prostate → axial bone (Batson's plexus), gut → liver (portal vein), kidney/sarcoma → lung (caval route).
- **Colonization-dominated regime.** Venous inflow is distributed across several comparable routes; subtype-specific molecular colonization programs become the driver. Example: breast, where the bone-versus-liver split tracks ER/HER2 subtype rather than anatomy.

Making the criterion non-circular. The framework is only meaningful if regime membership can be assigned *independently* of the metastatic outcome it is meant to explain; otherwise one classifies prostate as routing-dominated *because* it seeds bone and then claims prediction. We therefore define a dominance ratio D from anatomy alone:

$$D = \frac{\text{venous outflow fraction through the single largest first-pass bed}}{\text{summed outflow fraction through all other beds}},$$

computed from published venous-drainage anatomy of the primary site, with no reference to where metastases land. We propose (provisionally, pending calibration) that $D \gtrsim 2$ denotes the routing-dominated regime and $D \lesssim 1$ the colonization-dominated regime, with an explicit intermediate band. This threshold is currently a stipulation, not an empirically fitted value; fixing it against an anatomical dataset independent of outcome is required future work, and until then regime assignments near the boundary should be treated as provisional. Stating the criterion explicitly is what converts the framework from a post-hoc relabeling into a falsifiable one.

3 Model structure (a framing device, stated honestly)

The compartment calculation exists to instantiate the regime idea and generate predictions; it is not a fitted predictor, and its quantitative outputs are used only for ranking, never magnitude. We make its simplicity explicit rather than letting the differential-equation notation imply more.

For seven destination compartments (lung, liver, axial bone, appendicular bone, brain, adrenal, lymph node), burden M_o in organ o is written

$$\frac{dM_o}{dt} = f_o\left(1 - \frac{M_o}{K_o}\right) - c_o\,M_o,$$

with inflow f_o, capacity K_o, clearance c_o. *The compartments do not interact*, so each equation is independent and its stable steady state is the closed-form saturation value

$$M_o^* = \frac{f_o K_o}{f_o + c_o K_o}.$$

No integration is required; the dynamical notation is expository only. A coupled variant (compartments competing for a finite CTC pool) changed nothing material (breast bone:liver 1.65 → 1.52, no ranking change). The model therefore captures relative *delivery* and *retention*, and deliberately not inter-site competition, sequential (metastasis-to-metastasis) spread, dormancy, or CTC survival.

Inflow: three anatomical edge types, all set independently of the predicted tropism: (1) arterial perfusion (background); (2) venous first-pass routes—Batson's plexus → axial bone, portal vein → liver, caval route → lung, with Batson's valveless and pressure-gated; (3) barrier-crossing competence, a per-tumor gate (e.g. protease competence for the BBB), not an attraction term. An earlier hand-coded-affinity version was discarded as circular.

An acknowledged gap: lymphatic routing. The framework is venous-centric, yet the compartment set includes lymph node and clinical data show substantial nodal involvement (e.g. in breast). Lymphatic drainage is a third routing channel not represented in the mechanism; the lymph-node compartment is currently a passive sink rather than a modeled route. Extending the dominance criterion to lymphatic conduits is a needed generalization, not addressed here.

4 Evidence, and what it does and does not show

Ablation (within-model). Zeroing the venous first-pass terms drops top-site accuracy from 6/7 to 1/7, and all-correct from 76% to 0% across 3000 random parameter draws. This establishes that the venous terms carry the discriminating signal *within this model*—which is nearly forced, since the venous edges are the only per-cancer inputs (arterial perfusion, clearance, and capacity are shared). It is therefore an internal consistency check, not independent evidence that venous routing dominates *in vivo*; that stronger claim is tested for no cancer here and remains open (see §6).

Mechanism separation. Under ablation, prostate→bone and gut→liver collapse while melanoma→brain is unchanged (93%→93%): brain tropism depends on barrier-crossing competence, not routing, consistent with the brain having no venous first-pass shortcut.

Calibration. On the seven representable compartments, top-site is correct for 6/7 cancers; cosine similarity is 0.94 (prostate, autopsy) and 0.91 (gastric, SEER), computed on the representable subset and therefore optimistic in absolute terms. Ranking and ablation conclusions do not depend on these values.

5 The discriminating prediction and the marrow-volume confounder

The routing regime predicts that in Batson-routed prostate carcinoma, vertebral involvement declines monotonically from the lumbar entry point upward. The Bubendorf autopsy series ($n =$ 1,589) shows spine involvement falling ∼97% (lumbar) to ∼38% (cervical). Multiple retrograde-ascent decay rates fit this monotone pattern comparably; the claim is directional.

Countering the red-marrow-volume explanation. Vertebral metastasis might instead track hematopoietic (red) marrow volume, greater in the lumbar spine—a seed-and-soil alternative needing no retrograde flow. Three features argue against marrow volume as the *sole* driver. First, red-marrow fraction across vertebral levels does not fall as steeply or monotonically as the metastasis gradient. Second, and most cited as decisive, the gradient is *pressure-dependent*: in the classic Coman–DeLong experiments [4], raising intra-abdominal pressure drove lumbar vertebral metastases in ∼70% of animals versus essentially none without pressure, and marrow volume is unchanged by pressure. We flag that this keystone rests on mid-20th-century animal work whose modern replication we have not established; its weight should be discounted accordingly until reproduced. Third, Bubendorf report an inverse spine/lung relationship and earlier spine seeding (at smaller primary volumes), signatures of a distinct venous route. Marrow volume and retrograde venous pressure are not exclusive; the claim is only that the pressure route is needed to explain the full pattern.

A deeper confound: drainage and soil may be correlated. Venous first-pass beds and molecularly compatible niches need not be independent—a tumor that chronically seeds an organ may co-evolve receptors for it, so "routing" and "soil" could be two readings of one coupled process. If so, the two-regime distinction is a useful descriptive axis rather than a separation of independent causes. The dominance criterion D is defined from anatomy precisely to keep the routing axis measurable even if the two are entangled, but the independence of the regimes is an assumption, not an established fact.

6 Regime boundary: breast cancer, and an honest reading of P4

Breast is the model's one top-site failure, and the reason locates the regime boundary: bone-versus-liver tropism is governed by receptor subtype, and all breast subtypes share the same venous drainage, so routing cannot explain a subtype-dependent split.

What P4 does and does not establish. Using one SEER cohort [5] (Xiao et al., 2010–2013; population incidence, one denominator, all four subtypes, no estimated values), the bone:liver ratio varies 3.5-fold across subtypes (coefficient of variation 0.50), tracking the ER→bone / HER2→liver axis (ER-positive mean ratio 2.85 vs ER-negative 1.37; HER2-enriched most liver-shifted; Table 1). We are explicit about the logical status of this result: it confirms that breast bone-versus-liver is subtype-driven, which is the *mainstream* seed-and-soil expectation, and refutes only a routing-*only* account of breast that few would defend. P4 is therefore *consistency evidence that locates the colonization regime*, not a test of the framework's novel content. The genuinely novel claim—that venous routing *dominates* in prostate, gut, and kidney—has no comparably rigorous, outcome-independent test here and is the principal open item (§9). We state this asymmetry plainly: the paper's rigor currently sits where the stakes are lowest.

Breast subtype	Bone (%)	Liver (%)	Bone:Liver
HR+/HER2− (luminal A-like)	3.1	0.8	3.88
HR+/HER2+ (luminal B-like)	5.1	2.8	1.82
HR−/HER2+ (HER2-enriched)	4.6	4.2	1.10
HR−/HER2− (TNBC)	2.8	1.7	1.65

Table 1: Subtype-specific breast cancer metastasis and the bone:liver ratio, from a single SEER cohort (Xiao et al., *Oncotarget* 2017; SEER 2010–2013; population incidence rates). Venous drainage is identical across subtypes, so the 3.5-fold variation cannot be a routing effect; it tracks receptor subtype (ER-positive luminal-A most bone-shifted; HER2-enriched most liver-shifted), locating the colonization regime. All values are from the same cohort and denominator; none are estimated. HR, hormone receptor; HER2, human epidermal growth factor receptor 2; TNBC, triple-negative breast cancer.

7 Testable predictions

P1. Axial-bone concentration scales with venous-pressure exposure (Coman–DeLong, pending modern replication).

P2. Liver-versus-lung first-pass dominance follows the portal-versus-caval fraction of venous drainage.

P3. Brain tropism correlates with tumor protease/BBB-disruption capacity, not with any venous measure.

P4 (tested; consistency evidence).
Breast bone-versus-liver tracks ER/HER2 subtype, not anatomy (Table 1)—locating the colonization regime, not validating routing.

P5 (the key untested claim).
Across cancers, the anatomically-defined dominance ratio D predicts top metastatic site in the routing regime *independently of molecular markers.* This is the discriminating test the framework most needs and does not yet have.

8 Significance, generalization, and limitations

The transferable idea. Beyond metastasis, the reusable content is the decomposition of tissue localization into separable *arrival*, *retention*, and *colonization* operators, with different diseases (or tumor types) dominated by different operators. The same decomposition applies to infection tropism, embolic disease, and drug biodistribution. The regime criterion D is, moreover, *computable today* from existing venous-drainage anatomy with no new data, which makes regime assignment an immediately testable, low-cost hypothesis rather than a program requiring new measurement.

Translational hook. If the regime concept holds even directionally, it argues for stratifying metastasis-*prevention* trials by regime: anti-niche agents (bisphosphonates, denosumab, SCUBE2/Hedgehog-axis inhibitors) would be predicted to benefit colonization-regime tumors more than routing-regime tumors, reframing some regime-mismatched negative results as design artifacts rather than mechanism failures. This is a concrete, fundable study design, offered as a hypothesis.

Limitations. (i) The dominance threshold on D is stipulated, not fitted; near-boundary assignments are provisional. (ii) The compartment model is rank-only, uncoupled, and seven-node—tails (especially bone) are inflated, and aggregate breast over-predicts bone. (iii) The within-model ablation is near-forced and is not *in vivo* evidence. (iv) P4 confirms the mainstream subtype view, not the framework's novel routing claim, which is untested (P5). (v) Lymphatic routing is unmodeled. (vi) Routing and soil may be confounded rather than independent. (vii) Cross-dataset comparisons mix autopsy (end-stage cumulative) and SEER (clinically detected) data and are used for rank only; the P4 test avoids this by staying within one SEER cohort. (viii) Small sample (seven cancers). (ix) The reference list is limited to the primary anatomical and epidemiological sources directly supporting the argument [1–5]; a full submission would expand the molecular-mechanism citations (e.g. the SCUBE2 and CXCR4 literature) beyond the single subtype-pattern reference used here. The two-regime framing is a structured hypothesis, supported in part and not confirmed; no clinical use is implied.

References

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